# Unit 4 Review

*Solving Problems with Rational Numbers*

Name_____
Date_____
Period_____

*Show your work. Circle your final answer.*

*HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.*

## PART A. Adding and Subtracting in Scientific Notation

*Lesson 1 · 6 questions*

**1.** $(4.1 \times 10^{6}) + (2.3 \times 10^{6})$

**2.** $(7.6 \times 10^{-4}) - (2.9 \times 10^{-4})$

**3.** $(8.2 \times 10^{5}) + (4.9 \times 10^{5})$

**4.** $(5.4 \times 10^{8}) + (7.1 \times 10^{7})$

**5.** $(6.3 \times 10^{-2}) - (8.0 \times 10^{-4})$

**6.** Lena writes $(3.2\times10^4)+(5.5\times10^6)=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

## PART B. Multiplying and Dividing in Scientific Notation

*Lessons 2–3 · 6 questions*

**7.** $(4\times10^3)(2\times10^6)$

**8.** $\frac{7.2\times10^8}{1.8\times10^3}$

**9.** $(5\times10^6)(8\times10^{-2})$

**10.** $\frac{2.4\times10^5}{8\times10^9}$

**11.** $\frac{4\times10^7}{4\times10^{-3}}$

**12.** A laser emits $6.5 \times 10^{14}$ photons each second. How many does it emit in $4.0 \times 10^{2}$ seconds?

## PART C. Real-World Problems and Significant Digits

*Lesson 4 · 6 questions*

**13.** How many significant digits are in each measurement? 6,040 g · 0.00920 m · 45.00 s

**14.** A rectangular panel measures $3.2 \times 10^{2}$ mm by $1.45 \times 10^{2}$ mm. Find its area, with the correct number of significant digits.

**15.** A train travels $4.80 \times 10^{2}$ km in 6.0 hours. Find its average speed, with the correct number of significant digits.

**16.** Two lengths are measured: $5.40 \times 10^{3}$ mm and $2.8 \times 10^{2}$ mm. Find their sum, with the correct precision.

**17.** Two masses are measured: $7.60 \times 10^{-2}$ g and $3.42 \times 10^{-3}$ g. Find their difference, with the correct precision.

**18.** Micah adds two measured lengths, 2,340 m and 6.8 m, and says the sum must be written with two significant digits, as $2.3 \times 10^{3}$ m. Is he right? Explain, and give the correct sum.

## PART D. Evaluating Expressions with Radicals

*Lesson 5 · 7 questions*

**19.** $\sqrt{169}$

**20.** $\sqrt[3]{-8}$

**21.** $\sqrt{7^2 - 24}$

**22.** $\sqrt[3]{\frac{1}{27}}$

**23.** $\sqrt{100} \cdot 4^{-2}$

**24.** $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2 + 7}$

**25.** Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

## PART E. The Order of Operations

*Lesson 6 · 7 questions*

**26.** $24 - 3 \cdot 5 + 1$

**27.** $48 \div 6 \cdot 2$

**28.** $4 + 3(8 - 5)^2$

**29.** $\frac{6^2+9}{3^2-4}$

**30.** $-6^2+(-6)^2$

**31.** $20\left(\left(\frac{1}{2}\right)^2+5^{-1}\right)$

**32.** Which expression is equivalent in value to $2+4\cdot5^2$ ?

- ○ A. $900$
- ○ B. $102$
- ○ C. $42$
- ○ D. $150$

## PART F. Real-World Problems and the Order of Operations

*Lessons 7–8 · 5 questions*

**33.** Evaluate $5x^2$ when $x=-3$ .

**34.** Kinetic energy is $K=\frac{1}{2}mv^2$ . Find $K$ when $m=400$ kg and $v=7$ m/s.

**35.** A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

**36.** Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery.

**a.** What is the area of the patio?

**b.** Write an expression for the total cost.

**c.** Find the total cost.

**d.** A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

**37.** A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50 + 140)$
- ○ B. $18(3.50) + 140$
- ○ C. $18^2 + 3.50 + 140$
- ○ D. $18^2(3.50) + 140$